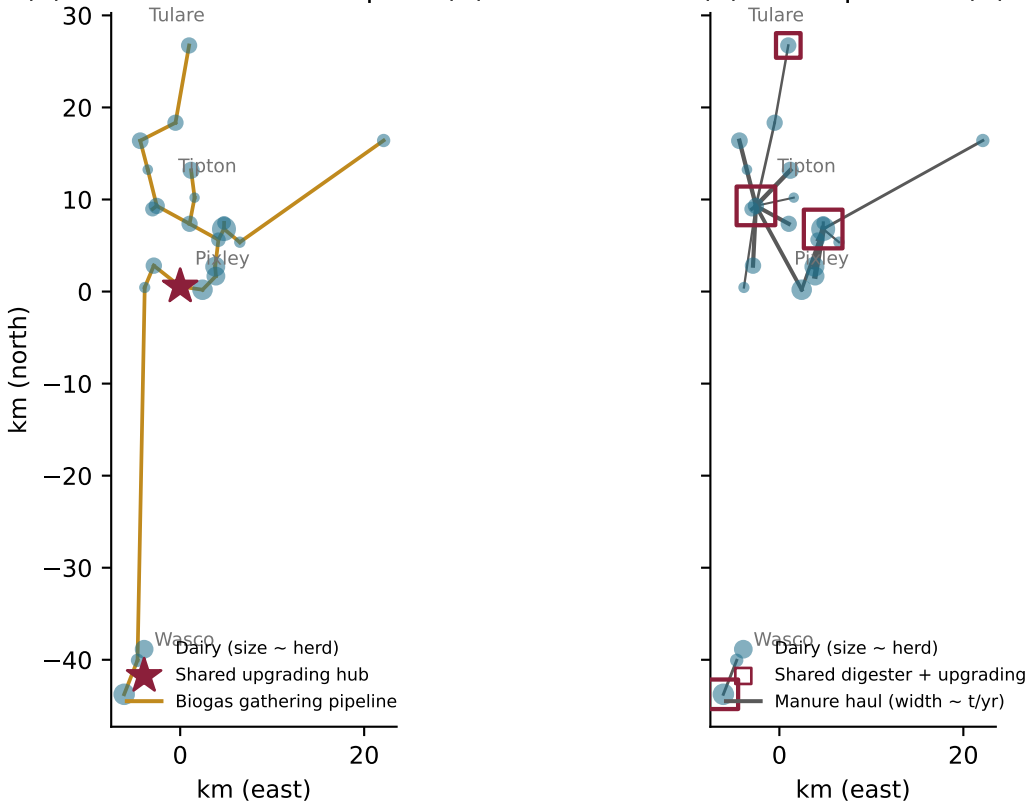
(a) Observed hub-and-spoke (B)
(b) MILP optimum (C)
Tulare
Tipton
Pixley
Wasco
km (north)
km (east)
30
20
10
0
−10
−20
−30
−40
0
20
Dairy (size ~ herd)
Shared upgrading hub
Biogas gathering pipeline
Shared digester + upgrading
Manure haul (width ~ t/yr)